Medical delivery quadcopter - component list

Design basis: 7 kg class quadcopter, 1 kg medkit, 3 km one way, 5 m/s wind. Power: 6S 18 Ah LiPo. Propulsion: Tarot 5008 340KV with 18 in props. Flight controller: Pixhawk 6C. Prepared for the hackathon submission.

1. Components, models and weights

ID	Item	Model / specification	Qty	Unit mass (g)	Total mass (g)	Basis
Propulsion						
P1	Brushless motor	Tarot TL96020 5008 340KV (14 poles, 58.5 mm dia., 25 mm 4-hole mount)	4	168	672	V
P2	Propeller	Tarot 1855-class 18 x 5.5 in carbon fibre, 2 CW + 2 CCW	4	25	100	E
P3	Prop adapter	Tarot quick-release prop adapter for 5008	4	12	48	E
P4	ESC	Hobbywing XRotor-40A (SKU 30901001), 2-6S, 40 A cont. / 60 A burst, no BEC	4	26	104	V
Power						
E1	Battery	6S1P 22.2 V, 18,000 mAh LiPo, 15C or higher, XT90-S (Tattu-class UAV pack)	1	2100	2100	E
E2	Main fuse	125 A slow-blow MIDI/ANL fuse with holder	1	30	30	E
E3	Power module + PDB	Holybro PM07 (14S): 90 A continuous, 140 A burst, dual 5.2 V outputs	1	47.5	47.5	V
E4	Servo BEC	5-6 V, 5 A UBEC (generic), separate servo rail	1	20	20	E
Avionics and communications						
A1	Flight controller	Holybro Pixhawk 6C, plastic case, 84.8 x 44 x 12.4 mm (ArduPilot Copter)	1	34.6	34.6	V
A2	Pixhawk cable set	Cables supplied with the 6C	1	15	15	E
A3	GPS + compass	Holybro M9N GPS (u-blox NEO-M9N, IST8310 compass), with case	1	32	32	V
A4	RC receiver	RadioMaster Bandit BR3 ExpressLRS 915/868 MHz, dual antenna	1	5.5	5.5	V
A5	Telemetry radio (air unit)	SiK-type telemetry radio, air side (model to be confirmed)	1	20	20	E
A6	Buzzer + strobe	5 V LED strobe and buzzer	1	20	20	E
A7	microSD card	32 GB high-endurance card for flight logs	1	1	1	E
Payload and safety						
L1	Release servo + latch	Metal-gear servo with hook latch (AUX OUT 1)	1	80	80	E
L2	Aircraft parachute	Recovery parachute system, ~7 kg class, servo release (AUX OUT 2)	1	500	500	E
L3	Medkit + cradle	1 kg payload, quick-release cradle on the CG axis	1	1000	1000	S
Structure						
S1	Centre plates	3 mm carbon fibre, about 260 mm across, 2 plates	1	350	350	E
S2	Arms	25 x 23 mm carbon tube, about 350 mm exposed, with clamps	4	112.5	450	E

ID	Item	Model / specification	Qty	Unit mass (g)	Total mass (g)	Basis
S3	Motor mounts	CNC aluminium plate, 25 mm 4-hole pattern	4	30	120	E
S4	Landing gear	Carbon tube legs, about 300 mm tall	1	250	250	E
S5	Fasteners	M3/M4 stainless, nylon-lock nuts, standoffs	1	130	130	E
Wiring and miscellaneous						
W1	Wires and connectors	8 AWG main lead, 14 AWG ESC feeds, XT90-S, XT60, 3.5 mm bullets, servo leads	1	200	200	E
W2	Miscellaneous	Heat shrink, cable ties, threadlocker	1	50	50	E

Basis column: V = weight taken from the manufacturer or seller listing; E = engineering estimate (replace with the datasheet value when the exact part is chosen); S = specified by the mission requirement.

2. Mass summary

Item	Mass
Aircraft without payload (dry)	5.38 kg
Medkit payload	1.00 kg
Takeoff mass	6.38 kg
Mass limit	7.00 kg
Margin to the limit	0.62 kg
Mass after the medkit is released	5.38 kg

Not counted in aircraft mass (ground equipment): RadioMaster Bandit ExpressLRS 915/868 MHz transmitter module (155 g with antenna), handheld transmitter, balance charger, laptop with Mission Planner.

3. Key performance results (formula sheet method, ISA sea level)

Quantity	Result
Takeoff mass	6.38 kg (limit 7.00 kg)
Thrust-to-weight	2.32 rated, 2.09 derated 90% (target 2.0)
Hover thrust per motor	15.6 N, 43% of rated 3.7 kgf
Hover power (nominal / with 15%% margin)	769 W / 880 W
Hover current and C-rate	41 A, 2.3C
Peak current, all four motors at max	122 A (6.8C); pack >= 15C = 270 A
Hover endurance, loaded (nominal / with margin)	23.7 min / 20.7 min (target 20 min)
Usable energy	304 Wh of 400 Wh (80% depth of discharge x 0.95)
Mission at 14 m/s: stated wind (head out, tail back)	12.7 min, 201 Wh, 34% left
Mission at 14 m/s: worst case (headwind both ways)	15.6 min, 246 Wh, 19% left, max one-way 2.95 km
Mission at 16 m/s: stated wind (head out, tail back)	11.4 min, 193 Wh, 37% left
Mission at 16 m/s: worst case (headwind both ways)	13.6 min, 230 Wh, 24% left, max one-way 3.22 km
PM07 loading at 16 m/s cruise	57 A of 90 A rated (64%)

Design rule: cruise at 16 m/s airspeed with the parachute kept. At 14 m/s the worst case leaves 19% of usable energy, just under the 20% reserve target. Results use assumed efficiencies (figure of merit 0.65, motor 0.85, ESC 0.95), a drag area of 0.09 m2 and a 25 W avionics load, so confirm them with manufacturer data before the final submission.

Open items to confirm: exact 18 Ah LiPo model and weight; telemetry radio and parachute models; frame mass from the final CAD.